Multiscale Lightweight Transformer Network With Adaptive Contrastive Learning for Hyperspectral Target Detection

Ying Cui, Tao Chen, Liguo Wang, *Member, IEEE*, Shan Gao, Liwei Chen, and Chunhui Zhao

***Abstract*—Hyperspectral target detection (HTD) faces significant challenges due to the scarcity of labeled samples and the redundancy in spectral information. To address these challenges, this article proposes a multiscale lightweight Transformer-based detector (MSLTD) specifically designed for HTD. First, a novel training sample generation strategy is developed that relies solely on raw hyperspectral data and a single prior target spectrum to generate a sufficient number of high-quality training samples. Second, a multiscale feature lightweight network based on a three-channel Transformer is proposed. It first divides spectral pixels into consecutive groups to extract local spectral features across adjacent bands and then introduces a multiscale sparse subspace Transformer (MSST) to capture and fuse intrinsic spectral characteristics and long-range dependencies at multiple resolutions while effectively reducing spectral redundancy. In addition, an adaptive hardness-aware contrastive (AHC) loss function is introduced, which integrates triplet loss and focal loss to adaptively emphasize hard-to-classify samples and improves the discriminative ability between targets and backgrounds. The effectiveness of the proposed MSLTD is validated through comparative experiments on four real hyperspectral datasets against seven state-of-the-art target detection methods, demonstrating improved detection accuracy and competitive computational efficiency. Extensive ablation studies further confirm the contributions of key components in the framework, highlighting its robustness and applicability in various HTD scenarios. The code for this work is available at https://github.com/2474109721ct/MSLTD for reproducibility purposes.**

***Index Terms*—Adaptive hardness-aware contrastive (AHC) loss function, hyperspectral target detection (HTD), lightweight, multiscale feature.**

I. Introduction

HYPERSPECTRAL target detection (HTD) is based on high-resolution spectral imaging technology, which detects targets by analyzing the spectral differences of different ground objects and has been widely used in many fields, such as agricultural monitoring [1], [2], [3], disaster warning [4], mineral exploration [5], pollution detection [6], food safety [7], medical diagnosis [8], and military reconnaissance [9].

In recent decades, scholars have developed many HTD algorithms.

Traditional hyperspectral target detectors typically rely on model-based assumptions and employ statistical techniques to construct decision functions. These methods focus primarily on accurately distinguishing targets from the background and generally fall into several categories, including spectral matching, statistical modeling, subspace projection, and sparse representation or energy minimization. Spectral matching was one of the first techniques applied to HTD, including the spectral angle mapper (SAM) [10] and the spectral information divergence (SID) [11]. The SAM detects targets by computing the angle between pixel spectra and a prior target spectrum, whereas SID identifies targets based on probabilistic divergences in spectral features. In contrast, statistical modeling approaches optimize detection by assuming that the background follows a known statistical distribution. Representative methods include the generalized likelihood ratio test (GLRT) [12], adaptive coherence estimator (ACE) [13], and the adaptive matched filter (AMF) [14], which use the principles of maximum likelihood estimation or error minimization. Subspace projection methods perform detection by projecting both the target and the background into a low-dimensional subspace and then discriminating between their representations. Examples include orthogonal subspace projection (OSP) [15] and matched subspace detector (MSD) [16]. In addition, sparse representation and energy minimization methods have advanced HTD techniques. For example, sparse target detection (STD) [17] represents pixel spectra using a learned dictionary and distinguishes targets based on reconstruction errors. Constrained energy minimization (CEM) [18] minimizes the output response of a finite impulse response filter

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Ying Cui, Tao Chen, Shan Gao, Liwei Chen, and Chunhui Zhao are with the College of Information and Communication Engineering, the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, and Heilongjiang Collaborative Innovation Center of Advanced Intelligent Perception Technology, Harbin Engineering University, Harbin 150001, China (e-mail: cuiying@hrbeu.edu.cn; chen-tao@hrbeu.edu.cn; gaoshan08@hrbeu.edu.cn; chenliwei@hrbeu.edu.cn; zhaochunhui@hrbeu.edu.cn).

Liguo Wang is with the College of Information and Communication Engineering, Dalian Minzu University, Dalian 116600, China (e-mail: wangliguo@hrbeu.edu.cn).

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by imposing custom-made constraints for target detection. Improved versions of CEM, such as hierarchical CEM (h-CEM) [19] and ensemble-based ECEM [20], have also been proposed to enhance detection performance. However, these traditional methods often rely on rigid statistical assumptions, which may not effectively reflect the intrinsic complexity of hyperspectral images in real-world scenarios, thus limiting their practical detection performance.

To overcome the limitations of conventional approaches, deep learning has demonstrated considerable efficacy and superiority in HTD, owing to its robust feature extraction capabilities and parallel computing advantages. Inspired by the Siamese network architecture, Zhu et al. [21] proposed a two-stream convolutional network target detector (TSCNTD), which processes paired target and background inputs using Siamese CNNs. The model subtracts the extracted features and then feeds them into fully connected layers to produce detection results. Zhu et al. [21] introduced a convolutional neural network-based detection (CNND) method that leverages labeled information from a source-domain hyperspectral dataset. In this approach, spectrally similar pixel pairs are labeled as class 0, while spectrally distinct pairs are labeled as class 1. These labeled pairs are subsequently used to train a multilayer CNN for binary classification in HTD tasks.

However, performance can degrade significantly when there is a mismatch between hyperspectral sensors in the source and target domains. To address this issue, Wang et al. [22] proposed a meta-learning-based HTD method, termed MLSN. By incorporating meta-learning principles, the meta-trained deep residual convolutional feature embedding (DRFE) module learns to distinguish spectral similarities and differences, enabling rapid adaptation to various HTD tasks. Although the aforementioned CNN-based detectors offer valuable insights for HTD, several limitations remain. First, typical CNN architectures involve multiple convolutional layers, resulting in substantial computational overhead. Second, when processing sequential spectral information, the inherently local receptive fields of 1-D CNNs limit their ability to capture global dependencies, which are critical for accurately interpreting spectral signals.

In recent years, Transformer-based architectures have gained increasing attention in HTD. Compared with convolutional neural networks (CNNs), Transformer-based methods demonstrate superior capability in capturing long-range dependencies along the spectral dimension, thereby enabling the extraction of more representative global features. For example, the Siamese Transformer target detector (STTD) [23] employs a dual-branch Transformer structure, in which high-purity pixels are mixed with prior spectral signatures to effectively model spectral sequence relationships. Building upon this foundation, the triplet spectral-wise Transformer detector (TSTTD) [24] introduces a triplet-based architecture along with the interclass separation and intraclass aggregation (ISIA) loss, which significantly improves the discriminative capacity of the learned feature space. The dual-branch Fourier Transformer target detector (DBFTTD) [25] incorporates prior frequency-domain information through a Fourier-based dual-branch design that replaces conventional multihead self-attention (MSA). In addition, learnable frequency-domain filters and a dynamic discard strategy are introduced, improving detection accuracy while enhancing computational efficiency. The momentum contrastive learning Transformer (MCLT) [26] employs a Transformer-based momentum encoder within an unsupervised learning framework. It leverages a memory queue mechanism for maintaining negative samples, setting a new benchmark for feature discrimination under label-scarce conditions. The compressive sensing Transformer for target detection (CS-TTD) [27] incorporates CS theory and enhanced global feature fusion strategies. It strengthens spectral feature discrimination by jointly optimizing computational efficiency and global context modeling in a Transformer framework, particularly under limited supervision. Despite these advancements, most existing Transformer-based HTD methods rely on full-spectrum inputs, which not only increase computational burden but also introduce vulnerability to interference from irrelevant or redundant spectral information.

Although Transformer-based deep Siamese networks have demonstrated promising capabilities in learning discriminative features and effectively separating targets from backgrounds, they generally require large volumes of high-quality training data [28]. However, in practice, only a single prior target spectrum is often available, which is insufficient to satisfy the data requirements of deep Siamese or triplet network architectures. To overcome this limitation, recent studies have explored the use of autoencoders and generative adversarial networks (GANs) to augment training samples [29], [30], [31], [32], [33]. While these generative methods offer potential benefits, the synthesized samples tend to closely resemble the original prior spectrum and fail to capture subtle spectral variations among target pixels. Moreover, they perform poorly in subpixel target detection scenarios and struggle with the inherent spectral variability commonly present in real-world hyperspectral data. In addition, existing Transformer-based models often encounter high computational complexity. Specifically, the multihead self-attention (MSA) mechanism involves computing interband relationships across all spectral channels, resulting in a quadratic increase in computational cost with respect to the number of spectral bands. This complexity significantly affects training efficiency and hinders scalability for large-scale hyperspectral datasets. From a loss function perspective, many methods formulate HTD as a binary classification task and adopt standard classification losses such as binary cross-entropy (BCE) [34], triplet loss [35], or focal loss [36]. Although these loss functions are effective for pixel-wise supervision, they fail to fully exploit the pairwise relationships encoded in the feature space. For example, BCE loss penalizes similarity scores between paired features but does not address class imbalance, which may introduce uncertainty and convergence issues during training. Triplet loss enhances feature separability by modeling relative distances among samples, yet it is sensitive to sample imbalance and scarcity, often resulting in unstable training dynamics. Focal loss focuses on difficult samples to mitigate imbalance, but it may excessively emphasize noisy or ambiguous instances, thereby reducing model robustness.

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Aiming at the challenges in existing methods, including sample scarcity, high computational complexity, and imbalanced sample distribution, we propose a multiscale lightweight Transformer-based detector (MSLTD) for HTD. The proposed method effectively addresses these key issues by constructing high-quality training samples using a spectral mixture model strategy, introducing a multiscale feature lightweight network based on a three-channel Transformer, and designing an adaptive hardness-aware contrastive (AHC) loss function that combines triplet loss with focal loss. The core innovations of this article are as follows.

1) A novel training sample generation method is introduced based on a spectral mixture model. This approach utilizes only the hyperspectral image and a single prior target spectrum to generate high-quality target samples. The K-means algorithm is applied to extract high-purity background endmembers, which are then linearly mixed with the prior target spectrum in random ratios to simulate subpixel targets. This strategy effectively addresses the issues of limited training samples and the challenge of subpixel target detection.
2) A multiscale feature lightweight network based on a three-channel Transformer is proposed to reduce spectral redundancy and enhance robust representation through a two-stage optimization framework. In the first stage, similarity modeling is used to generate interpretable weights and construct sparse feature vectors for key spectral band selection. In the second stage, a hierarchical architecture captures global dependencies at multiple resolutions, and cross-layer feature fusion further strengthens the expressiveness of the model.
3) An AHC loss function is developed to enhance feature discriminability. The loss jointly encourages the aggregation of positive samples with their anchors and the separation of negative samples, while adaptively adjusting the weights of hard samples to mitigate class imbalance. This contributes to more stable and effective model training in unbalanced detection scenarios.

II. Proposed Method

HTD aims to accurately identify and localize specific targets from complex scenes given a prior target spectrum. In this article, we propose the novel MSLTD detection method, the core process of which is described in detail below.

A. Overall Framework

The framework of the MSLTD is shown in Fig. 1. The training process consists of three main parts: training sample generation based on the spectral mixture model, a multiscale feature lightweight network based on a three-channel Transformer, and AHC loss. First, to overcome severe label scarcity, a large number of high-quality target training samples are generated by combining the to-be-measured hyperspectral images and a single prior target spectrum. Subsequently, the prior target spectrum, target samples, and background samples are fed into the lightweight multiscale feature network based on a three-channel Transformer to tackle spectral redundancy, which integrates the grouped spectral band embedding (GSBE) for local feature extraction and the multiscale sparse subspace Transformer (MSST) for global dependency modeling. Finally, to resolve class imbalance, the network is optimized by the AHC loss function to enhance target–background discrimination. Specific technical details will be presented in Section II.

B. Training Sample Generation Based on a Spectral Mixture Model Framework

HTD methods usually rely on a large number of high-quality training samples, and the network needs to learn features through the variability of sample pairs, while too many homogeneous samples may result in the model not being able to effectively learn the key features that distinguish the categories. In this article, we develop a method for training sample generation based on a spectral mixture model, as shown in Fig. 2. The linear mixing model (LMM) [23] assumes that each pixel in a hyperspectral image can be represented as a linear combination of multiple end elements and random noise. Based on this theoretical basis, we construct target training samples by mixing different types of background end-elements with a prior target spectrum in a random ratio.

First, we extract background samples from the input hyperspectral data matrix $X \in \mathbb{R}^{h\times w\times l}$, where l denotes the number of spectral bands and $n = h \times w$ represents the total pixels in the image. Using the CEM algorithm with a given prior target spectrum $\mathbf{x}^{\text{prior}} \in \mathbb{R}^{l\times 1}$, we generate a background spectral response map $D_c \in \mathbb{R}^{h\times w}$

$$D_c(i,j) = \frac{\hat{\mathbf{w}}^{\mathrm{T}}\mathbf{X}(i,j,:)}{\|\mathbf{X}(i,j,:)\|}, \quad i = 1,2,\ldots,h;\ j = 1,2,\ldots,w \tag{1}$$

where $\hat{\mathbf{w}}$ is the optimal filter and ε is a predefined threshold. The response map D_c is used to screen background samples by selecting pixels with values less than or equal to ε. The background sample matrix $\mathbf{X}_B$ is defined as

$$\mathbf{X}_B(i,j,:) = \begin{cases} \mathbf{X}(i,j,:), & \text{if } D_c(i,j) \le \varepsilon \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\mathbf{X}_B = [\mathbf{x}_1^b, \mathbf{x}_2^b, \ldots, \mathbf{x}_i^b, \ldots, \mathbf{x}_{n_B}^b] \in \mathbb{R}^{l\times n_B}$ denotes the extracted background sample set, l is the number of spectral bands, and n_B represents the total background samples.

To synthesize target samples, pure background endmembers are extracted from $\mathbf{X}_B$ using the K-means clustering method. By setting the cluster number to m, each cluster centroid corresponds to a background endmember. The background endmember matrix $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m] \in \mathbb{R}^{l\times m}$ is obtained by computing the mean spectrum of each cluster, where $\mathbf{u}_i \in \mathbb{R}^{l\times 1}$ represents the ith background endmember.

Following this process, we generate the target training sample set $\mathbf{X}_T = [\mathbf{x}_1^t, \mathbf{x}_2^t, \ldots, \mathbf{x}_{n_T}^t] \in \mathbb{R}^{l\times n_T}$ through linear mixing of prior target spectra $\mathbf{x}^{\text{prior}}$ and background endmembers $\mathbf{u}_i$:

$$\mathbf{x}_i^t = \lambda\mathbf{x}^{\text{prior}} + (1-\lambda)\mathbf{u}_i \tag{3}$$

where $\lambda \in [0.9, 1.0]$ controls the abundance ratio of the target spectrum. This range ensures the target proportion dominates

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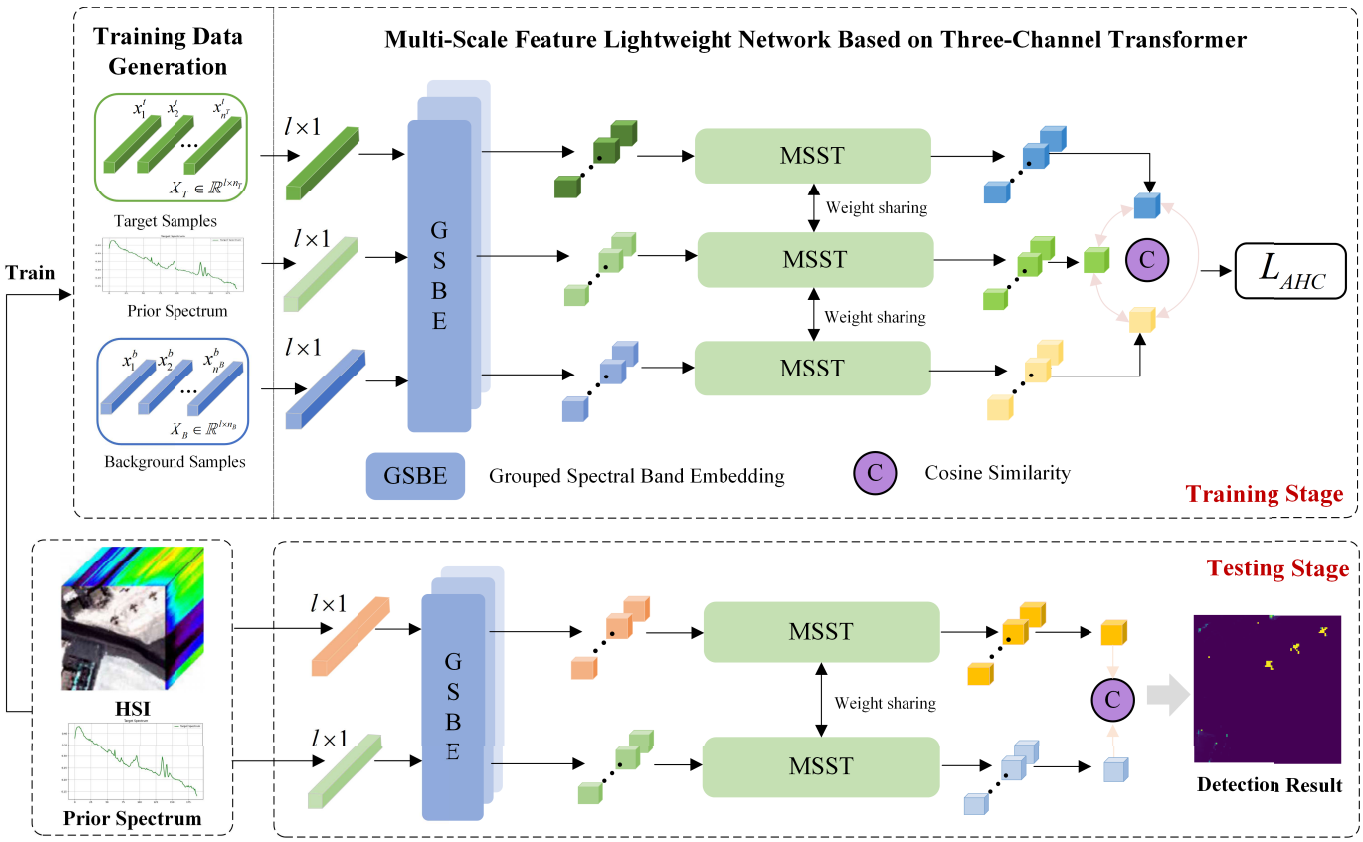


Fig. 1. Overview illustration of the proposed MSLTD for HTD.

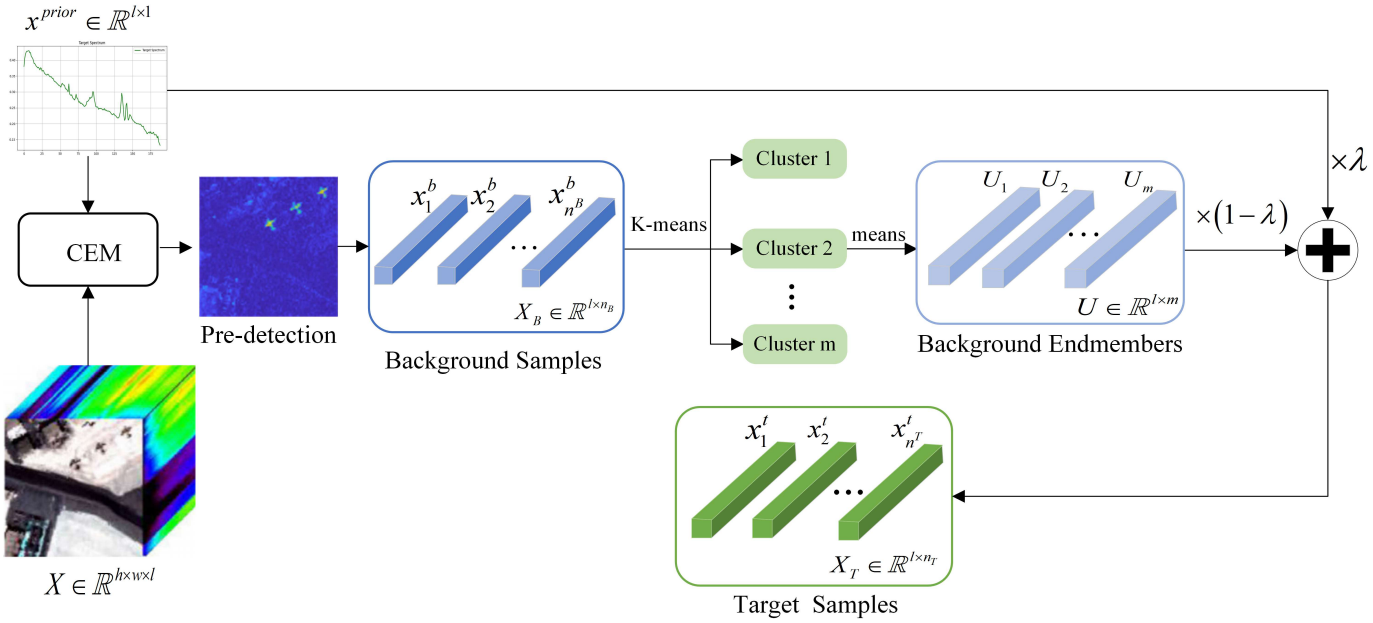


Fig. 2. Illustration of training sample generation in the proposed MSLTD.

in mixed pixels, as subpixel targets typically occupy less than 10% spatial coverage in hyperspectral imagery.

In this process, we only rely on a single a priori target spectrum and the hyperspectral image itself to obtain a large number of background samples and target samples as training samples, and in this article, we need to use the obtained training samples with the a priori target spectrum to construct two kinds of training sample pairs. In the training phase, we define the labels of the sample pairs composed of target samples and a prior target spectrum as 1, and the labels of the sample pairs

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composed of background samples and a prior spectrum as 0. In the testing phase, sample pairs are constructed by combining each pixel in the hyperspectral image to be detected with the a priori target spectrum and inputting these sample pairs into the trained network to obtain the detection score of each pixel. Ultimately, based on the detection scores, the target pixels can be effectively distinguished from the background pixels to achieve target detection.

C. Multiscale Feature Lightweight Network Based on a Three-Channel Transformer

This work transforms the target detection task into a spectral feature similarity discrimination problem and constructs a multiscale feature lightweight network based on a three-channel Transformer using shared parameters to achieve similarity metric learning. The network combines the GSBE and MSST. GSBE captures the spectral relationships between neighboring segments by dividing the spectrum into multiple spectral segments with fixed sequence lengths, thus enabling the MSST to efficiently extract multiresolution long-range spectral dependencies.

High-dimensional spectral data often exhibits significant local variability between adjacent bands [37], where these complex local spectral relationships are critical for effective discrimination of target pixels. Although traditional Vision Transformer (ViT) [38] patch partitioning strategies can reduce computational complexity, direct segmentation may lead to loss of spectral neighborhood information. To address this, we propose a dynamic spectral segmentation approach using 1-D convolution coupled with parametric local feature extraction. This method actively preserves discriminative differences between adjacent spectral segments while reducing Transformer sequence length, thereby effectively capturing key spectral variation features.

1) Grouped Spectral Band Embedding: Given an input pixel spectrum $\mathbf{X} \in \mathbb{R}^B$ with B spectral bands, the 1-D convolution operation is formulated as follows:

$$\mathbf{S}_X = \phi\left(\text{Conv1D}\left(\mathbf{X}; k, t, C\right)\right) \tag{4}$$

where Conv1D denotes the 1-D convolutional operation, k is the kernel length determining the spectral coverage per group, t represents the stride controlling intergroup spacing, C indicates the number of filters defining feature dimension, and $\phi(\cdot)$ is the nonlinear activation function.

This process transforms the input spectral data into grouped embeddings $\mathbf{S}_X \in \mathbb{R}^{G\times C}$, where the number of spectral groups G is calculated as follows:

$$G = \left\lfloor \frac{B-k}{t} \right\rfloor + 1. \tag{5}$$

2) Multiscale SST: The spectral embeddings generated by the GSBE can be treated as spectral sequences. To address the linear complexity challenges in capturing long-range dependencies and multiresolution analysis, this article proposes an MSST for extracting intrinsic spectral–temporal features across multiple resolutions. The structure of the MSST is shown in Fig. 3. The core module is the sparse subspace Transformer (SST), and the structure mainly consists of the self-low up path, the top-down path, and the transversely connected path. Sparse multiheaded subspace self-attention mechanism (SMSSM) from inside SST, which uses a fixed subspace basis $\hat{\mathbf{W}}$ to project the original spectral features into a low-dimensional subspace. This reduces the input dimensions of the attention computation and thus drastically reduces the computational complexity. The formula is

$$\mathbf{Q} = \mathbf{K} = \mathbf{V} = \hat{\mathbf{W}}. \tag{6}$$

The spectral dimension between K and Q^{T} is computed using matrix multiplication, resulting in the attention matrix M. Considering that the attention map may contain redundant features, it is prone to interference from irrelevant features. Therefore, we propose sparse subspace self-attention (SSA), which selects the top k elements with the highest weights from the attention matrix to sparsify the attention map. The remaining weights are set to zero, ensuring that attention is focused on the most important key features. This is specifically implemented using the following formula:

$$T_k\left(v_i\right) = \begin{cases} v_i, & \text{if } v_i \geq \min\left(\text{top-}k\right) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where the attention value v_i is reconstructed from the attention matrix M as a 1-D vector. The SSA mechanism reduces the complexity from $O((hw)^2 l)$ in MSA to $O(hwl^2/k)$ by selecting only the top k elements, significantly improving computational efficiency while retaining essential features.

The normalized matrix $T_k(A)$ after sparsification is then weighted and summed with V, generating the sparse self-attention output in subspace $\text{SSA}(Q, K, V) = A_{\text{sub_sp}} V$. In multihead mechanisms, applying sparse self-attention across multiple heads involves concatenating all head outputs through a linear transformation to produce the final output

$$\text{SMSSA}\left(Q, K, V\right) = \text{Concat}\left(\text{SSA}_1, \text{SSA}_2, \ldots, \text{SSA}_h\right) W_0 \tag{8}$$

where h represents the number of heads, Concat denotes concatenation of multihead outputs, and W_0 is the weight matrix for fully connected layers.

The input spectral sequence first undergoes root mean square normalization (RMSNorm), balancing the numerical range of spectral information and reducing variance drifts to enhance model training stability

$$\bar{X}_{i-1}^{j} = \frac{X_{i-1}^{j}}{\text{rms}\left(X_{i-1}\right)} \cdot h_{i-1}^{j} \tag{9}$$

where X_{i-1}^{j} indicates features at channel j of the input sequence to layer i, $\text{rms}(X_{i-1})$ signifies the root-mean-square value of the input sequence, and h_j are learnable parameters. Postnormalization sequences are transformed via linear transformation into two feature representations used for subsequent feature extraction and gating operations

$$Y_i = \text{Linear}_c\left(\bar{X}_{i-1}\right) \tag{10}$$

a convolution with a kernel size of 3 and a stride of 2 is applied to the spectral sequence to perform $2\times$ downsampling

$$Y_1^k = \text{Conv1D}\left(Y_1^{k-1}, 3, 2, 2^{k+1}C\right) \tag{11}$$

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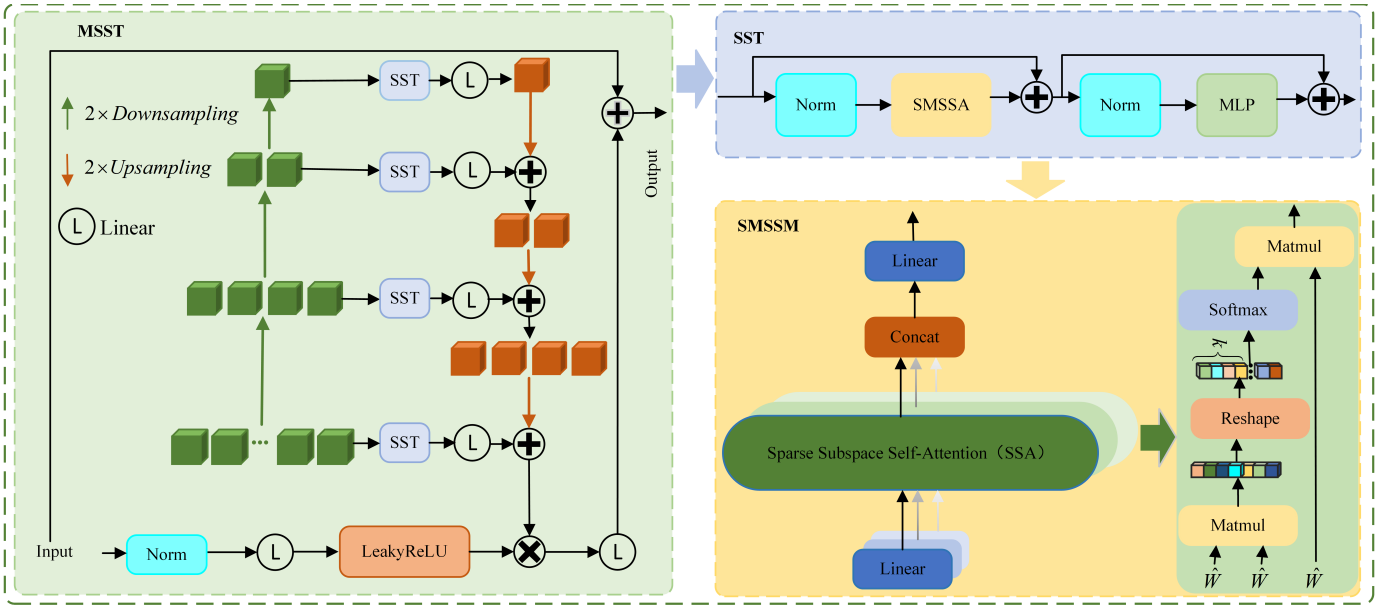


Fig. 3. Illustration of the MSST in the proposed MSLTD.

where $Y_1^0 = Y_1$ and $k \in \{1, 2, 3\}$ represents the kth layer of the multiscale structure. Each time downsampling is performed, the resolution is halved, and the number of channels is doubled. By extracting local spatial information through downsampling, multiple local features $\{Y_1^0, Y_1^1, Y_1^2, Y_1^3\}$ are formed, and these local features are enhanced through horizontal connection paths by SST to capture long-range dependencies between labels. This process can be represented as follows:

$$\hat{Y}_1^k = \mathrm{SST}\left(Y_1^k\right) \tag{12}$$

where $k \in \{0, 1, 2, 3\}$. From top to bottom path using a convolution kernel size of 3, a stride of 2, 1-D reverse convolution for low-resolution characteristic graphs for layered sampling, and through the SST enhanced feature map with top-down upsampling feature graph elements added, this process can be expressed as follows:

$$\tilde{Y}_1^3 = \mathrm{DConv1D}\left(\hat{Y}_1^3, 3, 2, 2^{3+1}C\right) \tag{13}$$

$$\tilde{Y}_1^k = \mathrm{DConv1D}\left(\left(\tilde{Y}_{1+1}^k + \mathrm{Linear}_{2^{k+1}}\tilde{Y}_1^k\right), 3, 2, 2^{k+1}C\right) \tag{14}$$

where $k \in \{2, 1, 0\}$, DConv1D denotes a transposed 1-D convolution with a kernel size of 3 and a stride of 2, used to perform $2\times$ upsampling. The fused feature map is projected onto the target dimension through a fully connected layer while filtering out redundant information. This process can be expressed as follows:

$$Y = \tilde{Y}_1^0 \otimes \mathrm{ReLu}\left(Y_1\right) \tag{15}$$

where $\otimes$ denotes element-wise multiplication. Finally, use a linear layer to adjust the label size and connect it with the input features via residual connections to reduce information loss and generate the final spectrum feature

$$X_t = X_{t-1} + \mathrm{Linear}_c\left(Y\right). \tag{16}$$

D. AHC Loss Function

The traditional triplet loss optimizes the relative distance between samples by minimizing intraclass differences and maximizing interclass separability, but it adopts an equal treatment strategy for all samples, making it difficult to effectively differentiate difficult samples located near the decision boundary. Although focal loss can alleviate the class imbalance problem by dynamically adjusting the sample weights, it may cause the training process to be unstable due to excessive attention to the difficult samples. To break through these limitations, this article proposes the AHC loss function, which can adaptively assign higher penalty weights to difficult samples while maintaining the overall balance of feature space optimization.

The core idea of the triplet loss function is to pull the features of a sample closer to the target sample of the same class while simultaneously pushing the target sample away from the background sample. Specifically, the mathematical definition of the triplet loss function is as follows:

$$L_{\mathrm{triplet}} = \frac{1}{N}\sum_{i=1}^{N}\max\left(0, \alpha + \cos\left(f_i^a, f_i^t\right) - \cos\left(f_i^a, f_i^f\right)\right) \tag{17}$$

where f_i^a is the anchor sample, which refers to the sample whose features are being compared, f_i^t is the positive sample, which is the target sample of the same class, f_i^f is the negative sample, which is the background sample, $\cos(\cdot)$ is the cosine similarity function, α is a hyperparameter representing the minimum margin between the negative sample and the anchor sample, and N is the number of samples.

The focal loss function is a dynamically re-weighted triplet loss function, which can help alleviate class imbalance and difficult classification problems. Its mathematical form is given by

$$L_{\mathrm{local}} = -\alpha\left((1 - p_t)^{\gamma}\right)\log\left(p_t\right) \tag{18}$$

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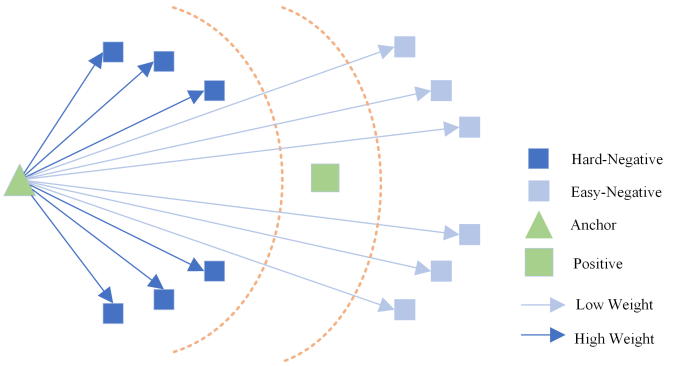


Fig. 4. Schematic of the AHC loss.

where p_t is the predicted probability for class t, α is the class weight factor, used to balance the negative sample importance, and γ is a hyperparameter that controls the difficulty of sample focus, adjusting the attention to harder samples.

To combine the advantages of triplet loss and focal loss, we modify the triplet loss function so that it works in conjunction with the focal loss function, dynamically increasing the weight of difficult samples in the classification. We use cosine similarity $\cos\left(f_i^a, f_i^t\right)$ and $\cos\left(f_i^a, f_i^f\right)$ to determine whether a sample is hard to classify. We experimentally determined the precise thresholds for hard-negative and easy-negative samples. Specifically, a sample is considered hard-negative if $\cos\left(f_i^a, f_i^t\right) > 0.7$ or $\cos\left(f_i^a, f_i^f\right) < 0.3$, and easy-negative if $\cos\left(f_i^a, f_i^t\right) < 0.3$ or $\cos\left(f_i^a, f_i^f\right) > 0.7$. For the anchor loss of positive samples, we define it as follows:

$$L_{\text{focal-p}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\alpha\left(1-\cos\left(f_i^a, f_i^t\right)\right)\log\left(\sigma\left(\cos\left(f_i^a, f_i^t\right)\right)\right)\right]. \tag{19}$$

For the anchor loss of negative samples, we define it as follows:

$$L_{\text{focal-n}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\alpha\left(1-\cos\left(f_i^a, f_i^f\right)\right)\log\left(\sigma\left(\cos\left(f_i^a, f_i^f\right)\right)\right)\right] \tag{20}$$

where $\sigma(\cdot)$ is the sigmoid function. Combining the advantages of triplet loss and focal loss, we define the AHC loss function as follows:

$$L_{\text{AHC}} = L_{\text{triplet}} + \frac{1}{2}\left(L_{\text{focal-p}} + L_{\text{focal-n}}\right). \tag{21}$$

In this formula, the coefficient 1/2 is experimentally tuned, ensuring the retention of the special learning ability of the triplet loss.

Moreover, it controls the degree of attention given to hard samples in the focal loss, preventing the model from training toward extreme samples, thus improving the stability and efficiency of the model. The illustration of the AHC loss is shown in Fig. 4.

III. Experimental Results and Analysis

A. Experimental Datasets

In our experiment, we used four hyperspectral images from two public hyperspectral datasets, as proposed in the method. The details are summarized as follows.

1) Sandiego Dataset: The first dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the Sandiego Airport in California, USA. The Sandiego dataset has a size of $100 \times 100 \times 224$, with a spatial resolution of 3.5 m, and the wavelength range from 400 to 2500 nm. The previous low-quality wavelength range (1–6, 33–35, 97, 107–113, 153–166, and 221–224) was excluded, leaving 189 wavelengths for experimentation [39]. The objective of the experiment is to detect a 3-D object, including 58 objects. The pseudo-color images of Sandiego with ground truth are shown in Fig. 5(a) and (b), respectively.

2) ABU Dataset [40]: The second dataset is the airport–beach–urban (ABU) dataset. The ABU dataset is mainly captured by AVIRIS sensors. It includes three types of scenes: airport, beach, and city. In addition, noise from the original image has been excluded. In the ABU dataset, we chose an airport scene image, a beach scene image, and an urban scene image as the representative datasets for the experiments, named Airport4, Beach1, and Urban2, respectively. Airport4 consists of 100 × 100 images, with 191 spectral bands. Beach1 has 150 × 150 images, with 188 spectral bands, and the spatial resolution is 4.4 m. Urban2 was captured over the coastal area of Southern California, with 100 × 100 images, containing 207 spectral bands, and the spatial resolution is 17.2 m. The pseudo-color images of Airport4, Beach1, and Urban2 with ground truth are shown in Figs. 6(a) and (b), 7(a) and (b), and 8(a) and (b), respectively.

B. Assessment Criteria

Multiangle performance validation of the proposed HTD method through both qualitative and quantitative evaluation metrics. The defined analysis is based on visual detection maps, subject-specific work (ROC curves [41]), and background–target distance-based evaluation of detection performance. The ROC curve maps the relationship between detection rate (P_d) and false detection rate (P_f) at different thresholds (τ) and evaluates detection performance through the ROC curve. A target detection effect improves as the target distance increases. The three 2-D curves [42] evaluate detection ability, target detection rate, and background suppression ability. The separability map [43] visualizes the distinction between targets and the background based on the distribution of detection scores. A larger distance between the target and background boxplots indicates better detection separability, while more compact box sizes reflect higher intraclass aggregation.

The defined analysis based on the area under the ROC curve (AUC [44]) is conducted. The commonly used AUC indicators include AUC(P_f, P_d), AUC(P_f, τ), and AUC(P_d, τ), which measure detection ability, adaptability, and background

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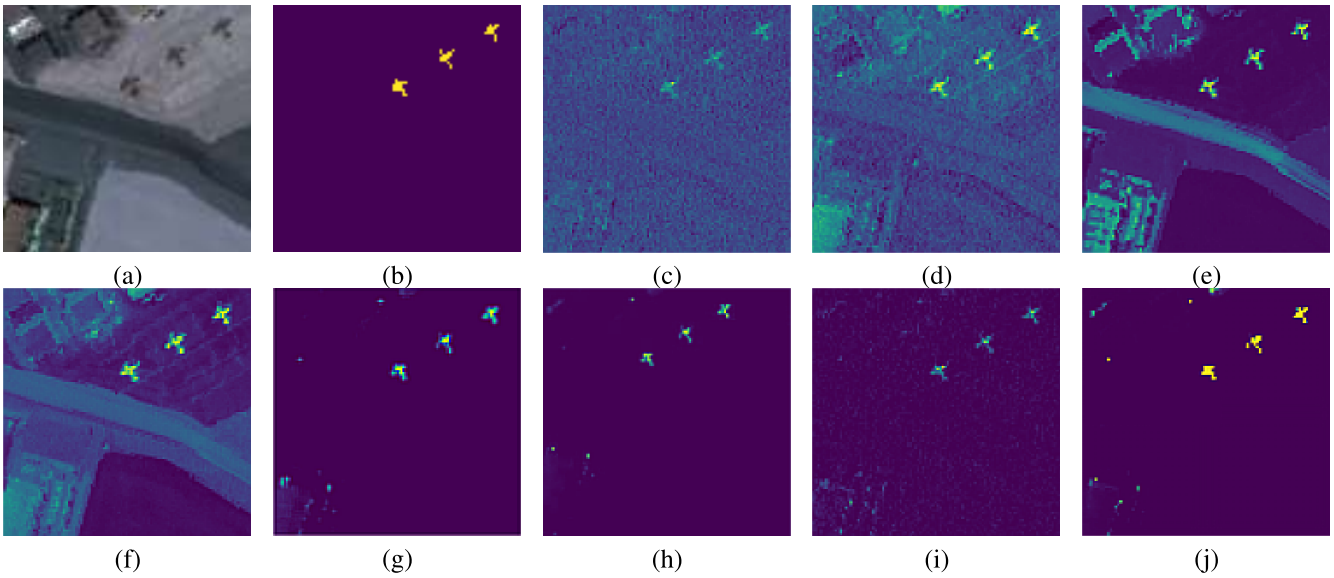


Fig. 5. Detection results of the Sandiego dataset. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) ECEM. (f) CNND. (g) HTD-IRN. (h) MCLT. (i) OS-VAE. (j) Ours.

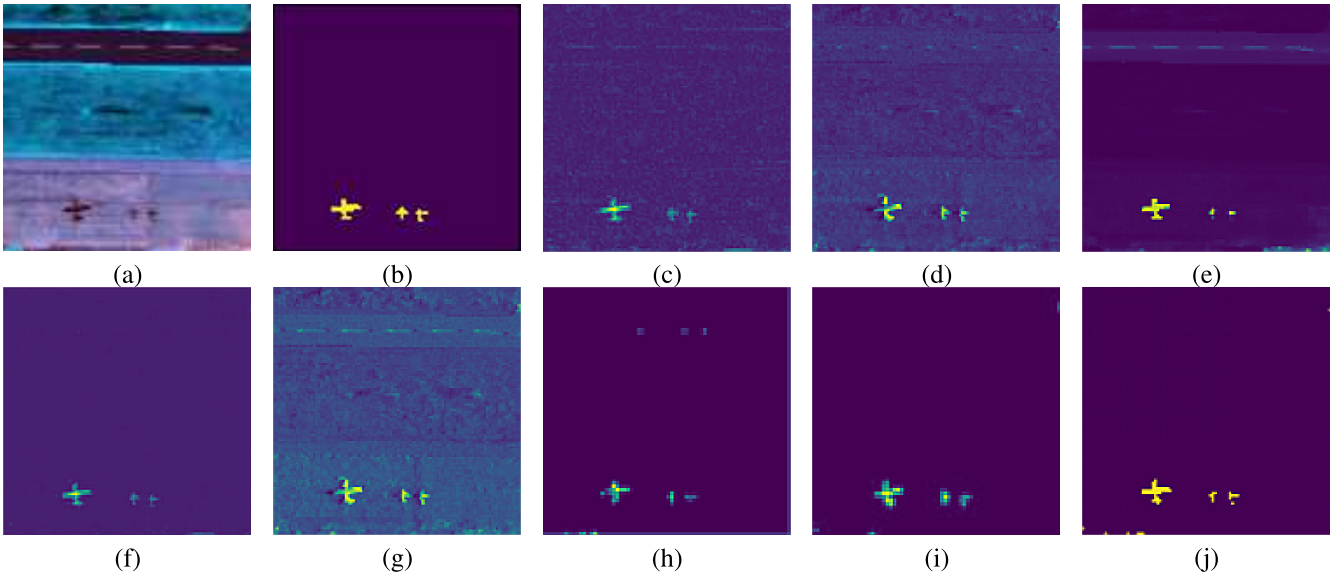


Fig. 6. Detection results of the Airport4 dataset. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) ECEM. (f) CNND. (g) HTD-IRN. (h) MCLT. (i) OS-VAE. (j) Ours.

suppression ability, respectively. To evaluate the overall performance, the following combined indicator is introduced:

$$\mathrm{AUC_{OA}} = \mathrm{AUC}\left(P_d, P_f\right) + \mathrm{AUC}\left(P_d, P_f\right) - \mathrm{AUC}\left(P_f, \tau\right). \quad (22)$$

At the same time, a comparison benchmark is defined

$$\mathrm{AUC_{SNPR}} = \frac{\mathrm{AUC}\left(P_d, \tau\right)}{\mathrm{AUC}\left(P_f, \tau\right)}. \quad (23)$$

Measures the balance between target detection performance and background suppression capability.

The detailed attributes of the above five indicators are listed in Table I.

TABLE I
PROPERTIES OF QUANTITATIVE METRICS

Metric	$\mathrm{AUC}(P_d, P_f)$	$\mathrm{AUC}(P_d, \tau)$	$\mathrm{AUC}(P_f, \tau)$	$\mathrm{AUC_{OA}}$	$\mathrm{AUC_{SNPR}}$
Perspective	Effectiveness	Detectability	False alarm	Overall	Overall
Range	[0, 1]	[0, 1]	[0, 1]	[-1, 2]	[0, ∞)
Trend	$\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\uparrow$
Optimum	1	1	0	2	∞

C. Experimental Setups

1) Comparison Method: The experiment demonstrates the effectiveness of the MSLTD method, comparing it with three classical methods and four advanced deep learning

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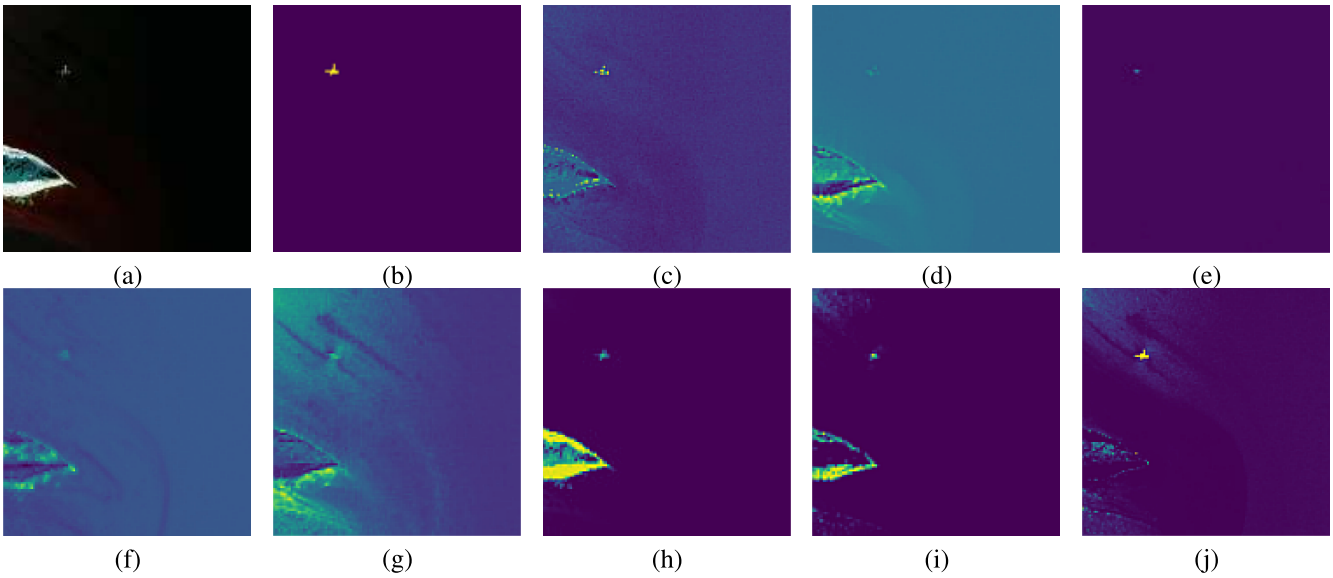


Fig. 7. Detection results of the Beach1 dataset. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) ECEM. (f) CNND. (g) HTD-IRN. (h) MCLT. (i) OS-VAE. (j) Ours.

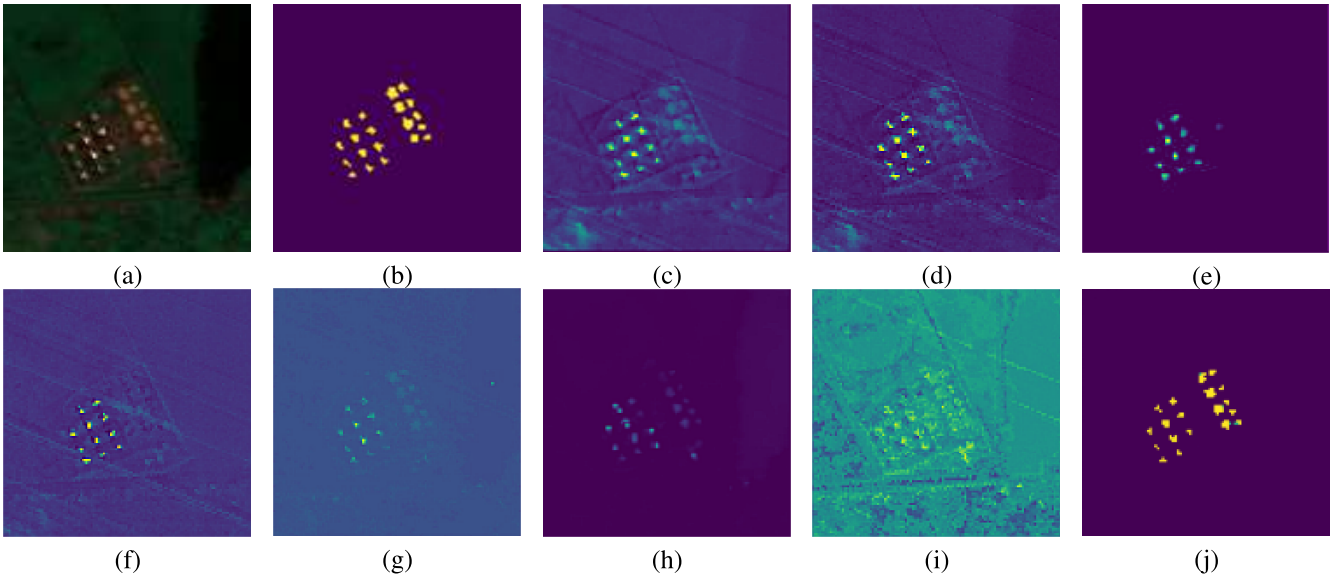


Fig. 8. Detection results of the Urban2 dataset. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) ECEM. (f) CNND. (g) HTD-IRN. (h) MCLT. (i) OS-VAE. (j) Ours.

methods. The classical methods mainly include CEM, OSP, and ECEM. The deep learning methods include CNND [21], HTD-IRN [45], MCLT [26], and OS-VAE [46]. For deep learning methods, the experimental configuration of each method is consistent with the setup described in the original paper. All methods are tested using the same target data to ensure a fair comparison. The MSLTD method uses Adam for optimization and sets a batch size of 64. The hyperparameters, such as learning rate and weight decay, are obtained through experimental tuning. The learning rate is finally set to 0.0001, and the weight decay parameter is set to 0. The ratio of the generated background training samples to the target training samples is 1:3, and this ratio is also determined through experiments. The hyperparameters α and γ in the focal loss are set to 0.5 and 2, respectively. The specific values are obtained through experimental optimization based on the performance of the model. All experiments are conducted in the PyTorch 2.0.1 environment, with the hardware configuration including an Intel Core i9-10900X CPU with a clock speed of 3.70 GHz, 32 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU.

2) Prior Target Spectrum: The prior target spectrum in the experiment is determined based on [47]. Specifically, the average spectrum of all target pixels is chosen as the prior target spectrum.

D. Results and Discussion

This section evaluates the effectiveness of the MSLTD through a series of comprehensive, multidimensional

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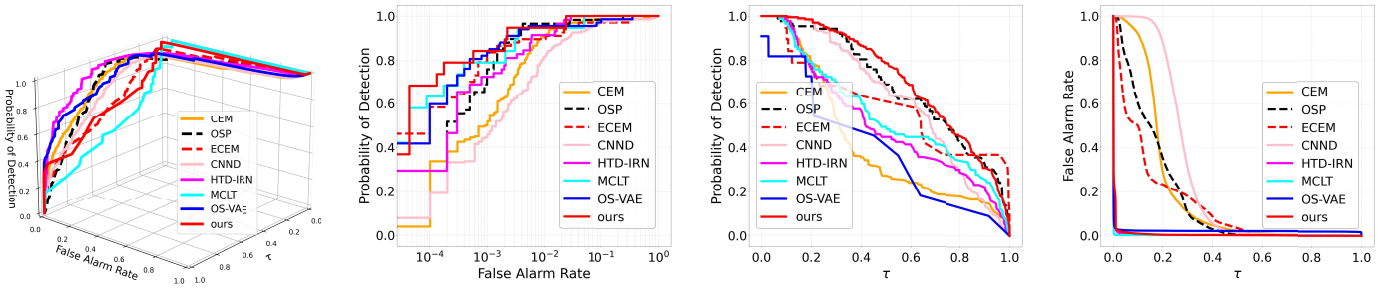


Fig. 9. 3-D ROC and the corresponding 2-D ROC curves of different methods for the Sandiego dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

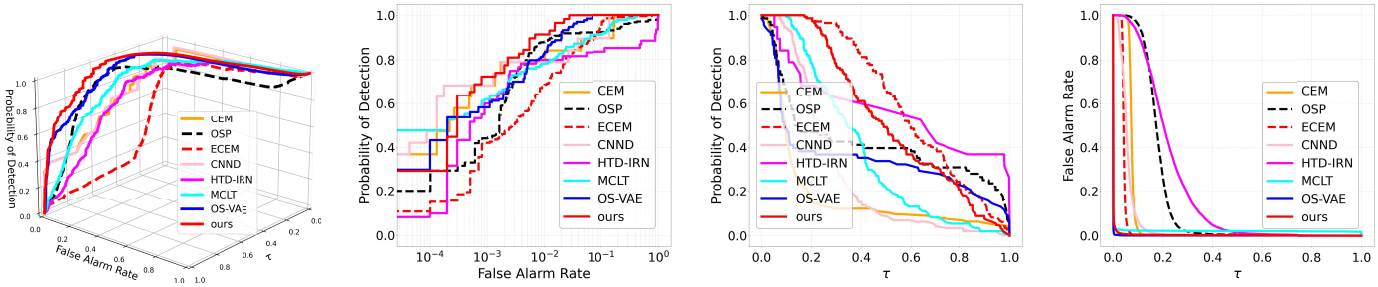


Fig. 10. 3-D ROC and the corresponding 2-D ROC curves of different methods for the Airport4 dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

experiments, including subjective visual inspection, ROC curve analysis, quantitative metric comparison, computational efficiency assessment, and target–background separability evaluation. The proposed method is benchmarked against seven state-of-the-art algorithms across four representative hyperspectral datasets.

Figs. 6(c)–(j)–8(c)–(j) present the detection maps of various methods across all datasets, while Fig. 5(c)–(j) focuses on the results for the San Diego dataset. Traditional algorithms such as CEM and OSP are capable of identifying targets, but they exhibit substantial background residuals and indistinct target boundaries. Although ECEM enhances background suppression, it suffers from fragmented target regions. Among learning-based detectors, CNND and HTD-IRN improve target integrity over traditional methods but still suffer from isolated high-response areas within the background. MCLT and OS-VAE demonstrate superior background suppression via global feature modeling; however, their sensitivity to small or subtle targets is insufficient, often resulting in missed detections. In contrast, the MSLTD maintains the complete geometric structure of all three aircraft targets and exhibits nearly zero-background response, producing detection maps that closely align with the ground truth. This superior performance is primarily attributed to the MSST module, which effectively captures multiscale spectral dependencies, suppresses redundancy, and enhances global spectral representation, thereby enabling robust target discrimination in complex scenes. As illustrated in Fig. 6(c)–(j), the MSLTD accurately localizes aircraft in the airport scene with negligible false positives, whereas CEM produces numerous false alarms in reflective regions. MCLT, despite its strong global modeling capacity, underperforms in delineating target boundaries. Fig. 7(c)–(j) demonstrates that the MSLTD achieves high-fidelity target detail capture, whereas CEM erroneously identifies sea foam as targets, and HTD-IRN exhibits detection failure for the ship stern region under complex illumination conditions. Although OS-VAE excels at eliminating background interference, it also sacrifices considerable target information. Finally, Fig. 8(c)–(j) highlights that the MSLTD achieves precise target localization and complete background suppression in challenging environments, whereas ECEM and OS-VAE produce false alarms in shadowed regions.

Figs. 9–12 present the 3-D ROC surfaces and the corresponding 2-D ROC curves for the MSLTD and seven comparison methods across four hyperspectral datasets. The 3-D ROC curves comprehensively characterize the tradeoff between detection P_d, P_f, and τ in a unified visualization framework. The 2-D ROC curves are evaluated under three common projections: ROC(P_d, P_f), ROC(P_f, τ), and ROC(P_d, τ), where P_d denotes the probability of detection, P_f denotes the probability of false alarm, and τ denotes the detection threshold. In the ROC(P_d, P_f) projection, the closer the curve is to the upper-left corner, the better the detection performance. As observed in Figs. 9–12, the MSLTD consistently achieves high detection rates with low false alarm probabilities across all datasets, reflecting its strong detection capability. For the ROC(P_f, τ) projection, curves closer to the lower-left corner indicate better robustness against false alarms under varying threshold settings. The MSLTD exhibits favorable performance in this regard, with curves consistently concentrated near the

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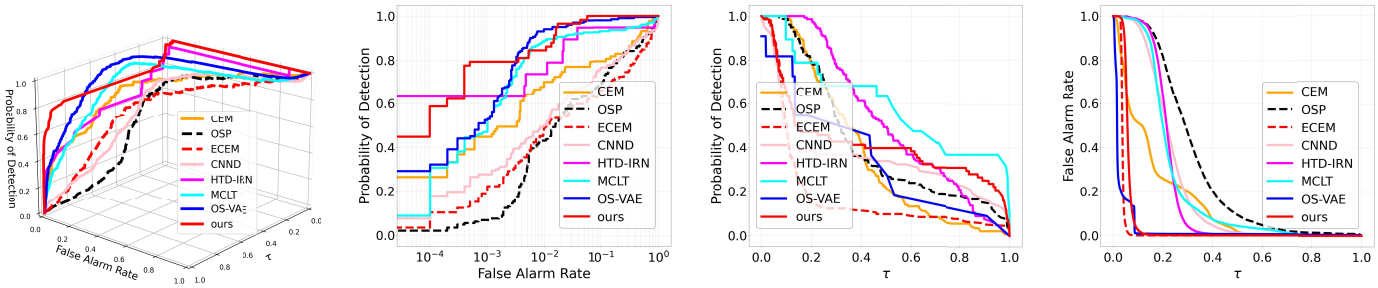


Fig. 11. 3-D ROC and the corresponding 2-D ROC curves of different methods for the Beach1 dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

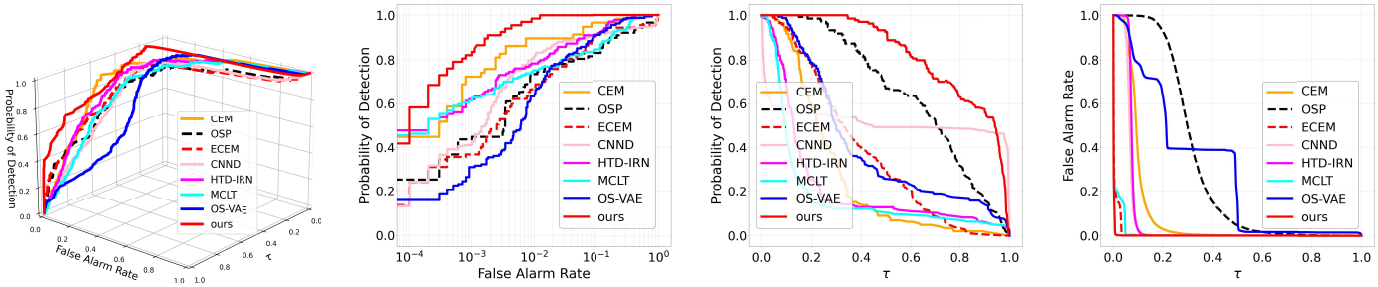


Fig. 12. 3-D ROC and the corresponding 2-D ROC curves of different methods for the Urban2 dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (PD, PF). (c) 2-D ROC curve of (PD, τ). (d) 2-D ROC curve of (PF, τ).

optimal region. The MSLTD shows clear advantages in the ROC(P_d, τ) projection. Its curve stays closer to the upper-left corner of the plot, which reflects strong background suppression under various detection thresholds. Compared with ECEM, the MSLTD achieves a better balance between sensitivity and specificity, offering more reliable detection performance. In summary, the MSLTD demonstrates the most competitive performance in both detection accuracy and background suppression ability across all ROC-based evaluations.

Although ROC curves offer valuable insights into detection performance, their interpretability can be limited due to the significant variation in detection behaviors among different methods. To enable a more comprehensive and quantitative evaluation, we report five AUC-based metrics and the corresponding inference times in Table II. The best-performing results are highlighted, followed by rankings of the second-best scores for each dataset. On all four datasets, the MSLTD achieves the best performance in AUC(P_d, P_f), outperforming the second-best results by 0.0099, 0.00312, 0.00755, and 0.02191, respectively. Although the MSLTD does not lead in AUC(P_f, τ), it consistently outperforms competing methods in AUC(P_d, τ), AUC_{oa}, and AUC_{SNPR}, reflecting its balanced performance across multiple dimensions of detection effectiveness. MCLT and OS-VAE achieve the best AUC(P_f, τ) scores on the San Diego, Airport4, and Beach1 datasets; however, their performance on AUC(P_d, τ) is suboptimal, indicating limited robustness under varying detection thresholds. In terms of computational efficiency, Table II shows that CEM and OSP require the least runtime due to their algorithmic simplicity. Nonetheless, the MSLTD achieves a favorable tradeoff between accuracy and efficiency. Benefiting from the design of the SSA module, which reduces the computational complexity to $O(hwl^2/k)$, the MSLTD achieves competitive inference speed, significantly outperforming OS-VAE and CNND during runtime, while still maintaining superior detection performance. This balance between high accuracy and real-time processing power enables the MSLTD to address real-world challenges in time-sensitive applications.

The separability box plot further validates that the proposed MSLTD method demonstrates strong capabilities in both background suppression and target enhancement across all four hyperspectral datasets. As illustrated in Fig. 13, the box plots visualize the separability between target and background responses for the MSLTD and the comparison methods. To reduce the influence of extreme values, the top and bottom 10% of pixel responses in each class are excluded. In the plots, the red boxes represent the statistical distribution of target responses, while the green boxes correspond to background responses. The central line in each box denotes the median, and the upper and lower whiskers indicate the maximum and minimum values, respectively. It is evident from Fig. 13 that the MSLTD yields a substantial interclass margin and minimal overlap between the target and background distributions, reflecting high feature discriminability. While MCLT achieves slightly stronger background suppression, it lags behind the MSLTD in overall target–background separability, particularly in maintaining distinct boundaries between the two classes.

In summary, the MSLTD exhibits superior performance in detection visualization, quantitative metrics, and separability analysis. Its robustness is especially pronounced in

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON FOUR HSI DATASETS

HSI Data Sets		Methods							
		CEM	OSP	ECEM	CNND	HTD-IRN	MCLT	OS-VAE	Ours
Sandiego	AUC(P_r, P_d)	0.9869	0.99445	0.98644	0.97687	0.99746	0.99684	0.99123	**0.99845**
	AUC(P_r, r)	0.19453	0.162	0.13709	0.27673	0.00344	0.00244	0.01938	0.00552
	AUC(P_d, r)	0.42513	0.68624	0.5945	0.63061	0.51332	0.55265	0.39698	**0.71507**
	AUC_{OA}	1.2175	1.51869	1.44385	1.33075	1.50734	1.54705	1.36883	**1.708**
	AUC_{SNPR}	2.18542	4.23604	4.33656	2.27879	149.221	**226.495**	20.4841	129.541
	Time(s)	**0.0465**	0.0816	4.3126	10.56715	0.25691	0.9394	2.56798	0.86548
Airport4	AUC(P_r, P_d)	0.97871	0.95965	0.97722	0.98127	0.86871	0.97719	0.99423	**0.99744**
	AUC(P_r, r)	0.075	0.17697	0.0437	0.05747	0.21861	0.02151	**0.00041**	0.00344
	AUC(P_d, r)	0.18374	0.4176	**0.60199**	0.27709	0.57079	0.37809	0.34236	0.54324
	AUC_{OA}	1.08745	1.20028	1.53551	1.20089	1.22089	1.33377	1.33618	**1.53724**
	AUC_{SNPR}	2.44986	2.35972	13.7755	4.82147	2.61099	17.5774	**835.041**	157.918
	Time(s)	**0.04672**	0.08653	4.6732	11.69273	0.21766	1.0905	2.62658	0.91746
Beach1	AUC(P_r, P_d)	0.90084	0.85331	0.81166	0.86333	0.94883	0.95333	0.98929	**0.99684**
	AUC(P_r, r)	0.15296	0.31518	0.03771	0.22612	0.21636	**0.02475**	0.03112	0.06038
	AUC(P_d, r)	0.37809	0.42477	0.18483	0.34848	0.53338	0.59807	0.34731	**0.8997**
	AUC_{OA}	1.12597	0.9629	0.95878	0.98569	1.26585	1.52665	1.30548	**1.83616**
	AUC_{SNPR}	2.47182	1.34771	4.90135	1.54112	2.46524	**24.1644**	11.1603	14.9006
	Time(s)	**0.112**	0.19352	9.87392	23.9363	0.5839	2.7351	5.9638	1.74792
Urban2	AUC(P_r, P_d)	0.83245	0.925	0.93399	0.93642	0.97719	0.95543	0.96472	**0.9991**
	AUC(P_r, r)	0.09863	0.31559	0.00672	0.06748	0.0768	**0.00926**	0.29097	0.00191
	AUC(P_d, r)	0.27056	0.65778	0.3772	0.51628	0.22066	0.19312	0.41275	**0.82627**
	AUC_{OA}	1.00438	1.26719	1.30447	1.38522	1.12105	1.13929	1.0865	**1.82346**
	AUC_{SNPR}	2.74318	2.08428	56.1309	7.65085	2.87317	20.8552	1.41853	**432.602**
	Time(s)	**0.05749**	0.09732	5.3913	12.1558	0.2794	1.1385	2.7389	1.02835

The best and second-best results are bolded and underlined, respectively.

complex scenes involving cluttered backgrounds and small, low-contrast targets. The effectiveness of the MSLTD is largely attributed to the integration of the MSST module and the AHC loss function, which together enhance global spectral representation, suppress irrelevant background information, and improve the detectability of small-scale targets.

E. Ablation Studies

This section conducts ablation experiments on four datasets to verify the effectiveness and necessity of the spectral mixture model, the SSA module, and the AHC loss function in the MSLTD framework. We designed eight network variants and evaluated them using a controlled variable approach, where all configurations were kept identical except for the ablated components. Specifically, Variant 1 combines multidimensional convolutional head transposed self-attention (MDTA) [48] with triplet loss to analyze the effect of simultaneously replacing the attention mechanism and the loss function; Variant 2 is based on the LMM and SSA but replaces AHC with triplet loss as another control experiment; Variant 3 integrates MDTA with AHC under the LMM framework; Variant 4 incorporates SSA with AHC under the LMM to validate the importance of the spectral mixture-based training sample synthesis method; Variant 5 employs SSA and the spectral mixture model but removes AHC and adopts triplet loss to examine the contribution of AHC in distinguishing hard samples; Variants 6–8 are constructed under the spectral mixture model with AHC but employ different attention mechanisms, to compare their relative impacts. For the first five variants, only AUC(P_f, P_d) results are reported, while the last three variants are evaluated in terms of AUC(P_f, P_d), parameter count (Params), and computational complexity (FLOPs). The results of all methods are summarized in Table III. In addition, we conduct a sensitivity analysis on the sparsity control parameter k in the SSA module, with detailed results presented in Fig. 14.

1) Ablation Study on Core Component Substitutions: We design Variants 1–3, mainly to examine the interactions between different modules. Specifically, Variant 1 keeps the spectral-mixture generator intact but replaces SSA with MDTA and substitutes AHC with a plain triplet loss to assess the contribution of the generator alone when both attention and loss designs are weakened. Variant 2 retains SSA and the triplet loss but reverts the generator to the conventional LMM, thereby evaluating the effect of SSA under a weaker sample generation process. Variant 3 adopts the weaker LMM

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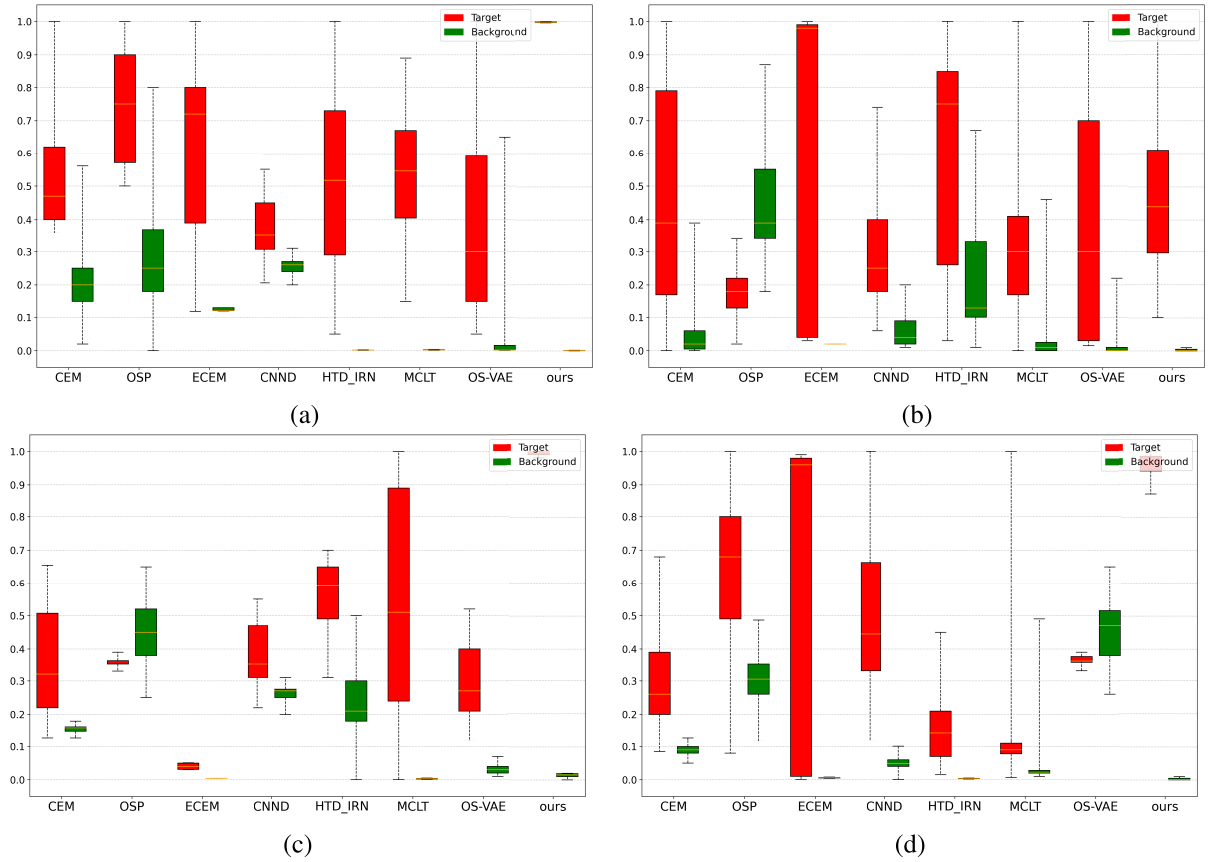


Fig. 13. Separability map. (a) Sandiego. (b) Airport4. (c) Beach1. (d) Urban2.

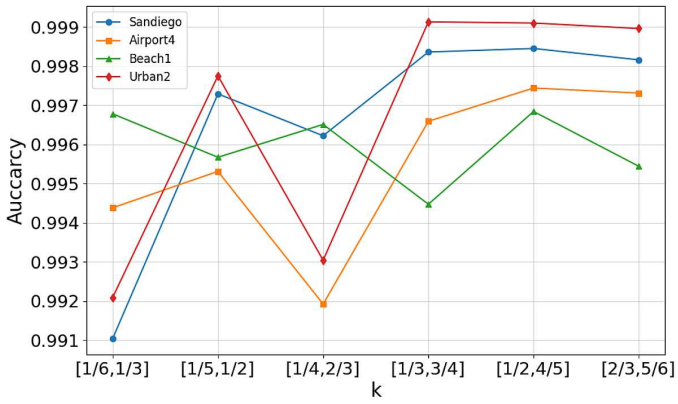


Fig. 14. Ablation qualitative comparison for different number k in the SSA.

generator together with MDTA attention but still employs the AHC loss, allowing the independent impact of AHC to be measured. Joint inspection of Variants 1–3 reveals that no single module works optimally in isolation: the spectral-mixture generator enhances the representational capacity of SSA, AHC stabilizes training across different generator and attention designs, and SSA compensates for the limited global modeling of MDTA. Thus, it is the cumulative interaction of these modules—rather than any individual component—that drives the superior performance of the MSLTD framework.

The Variant 4 employs an LMM [23] instead of the spectral mixing model to construct training samples, while keeping all other components unchanged. In this approach, the LMM treats every pixel in the hyperspectral image under test as a background sample and directly mixes these background samples with target samples to generate target pixels. As shown in Table III, the detection performance of this variant is inferior to the proposed method. Further analysis reveals that the LMM fails to effectively simulate subtle spectral differences among target pixels, while also exhibiting limitations in handling subpixel targets and spectral variability issues.

Variant 5 replaces the proposed AHC with conventional triplet loss, while keeping all other components unchanged. The triplet loss assigns uniform weights to all training samples regardless of their class proportions or discrimination difficulty. As shown in Table III, this variant yields inferior detection results across all four datasets compared to our proposed method. In contrast to AHC loss, the triplet-based variants fail to capture numerous hard-to-classify targets, resulting in incomplete preservation of morphological information for detected objects. This limitation significantly degrades the structural integrity and completeness of detection outputs.

Variants 6–8 adopt three different self-attention mechanisms, namely top-K sparse attention (TKSA) [49], random sparse self-attention (RSSA) [50], and MDTA [48], to replace the SSA, while keeping all other components unchanged. As shown in Table III, TKSA reduces computational cost through

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TABLE III
QUANTITATIVE ANALYSIS OF AUC, PARAMETERS, AND FLOPS FOR MSLTD VARIANTS IN ABLATION EXPERIMENTS

Methods	Indexes	Sandiego	Airport4	Beach1	Urban2
Spectral Mixture Model+MDTA+triplet loss (Variant 1)	AUC(P_f, P_d)	0.96021	0.984223	0.9559	0.98647
LMM+SSA+triplet loss (Variant 2)	AUC(P_f, P_d)	0.99068	0.99317	0.99367	0.99581
LMM+MDTA+AHC (Variant 3)	AUC(P_f, P_d)	0.96533	0.98509	0.9542	0.98673
LMM+SSA+AHC (Variant 4)	AUC(P_f, P_d)	0.99074	0.99413	0.99379	0.99681
Spectral Mixture Model+SSA+triplet loss (Variant 5)	AUC(P_f, P_d)	0.96741	0.98522	0.95600	0.98688
Spectral Mixture Model+TKSA+AHC (Variant 6)	AUC(P_f, P_d)	0.99614	0.99492	0.99407	0.99786
	Params (M)	0.131	0.131	0.131	0.131
	FLOPs (G)	0.605	**0.621**	1.503	0.684
Spectral Mixture Model+RSSA+AHC (Variant 7)	AUC(P_f, P_d)	0.99684	0.99658	0.99529	0.99682
	Params (M)	0.133	0.133	0.133	0.133
	FLOPs (G)	0.613	0.629	1.498	0.712
Spectral Mixture Model+MDTA+AHC (Variant 8)	AUC(P_f, P_d)	0.99431	0.99637	0.99513	0.99831
	Params (M)	0.134	0.134	0.134	0.134
	FLOPs (G)	0.615	0.631	1.517	0.682
Proposed MSLTD	AUC(P_f, P_d)	**0.99845**	**0.99744**	**0.99684**	**0.9991**
	Params (M)	**0.130**	**0.130**	**0.130**	**0.130**
	FLOPs (G)	**0.601**	0.622	**1.443**	**0.666**

The bold represents the best result.

a fixed top-k selection strategy, but its static sparsity pattern limits adaptability to dynamic spectral variations, resulting in slightly lower detection performance. RSSA introduces random sparsity to improve efficiency, yet its inherent instability often leads to the omission of critical features. MDTA enhances local feature representation via convolutional heads, but its global modeling capacity remains limited. In contrast, SSA achieves superior performance across all datasets while maintaining lower parameter and FLOPs budgets. By combining sparse attention allocation with subspace projection, it effectively preserves global spectral dependencies while suppressing redundancy, demonstrating an excellent balance between accuracy and efficiency.

2) Impact of Parameter k: We investigate the influence of the critical parameter k in SSA, with detailed analysis shown in Fig. 14. The optimal sparsity level is determined by the appropriate selection of k. To avoid exhaustive parameter search, we define a controlled interval range for k to dynamically identify the most contributive attention scores. Experimental results demonstrate that when k is too small, the performance significantly degrades due to insufficient aggregation of global contextual information. By constraining k within the range $[1/2, 4/5]$ of the total spectral bands, SSA achieves optimal detection accuracy across all four hyperspectral datasets. However, further increasing k beyond this interval progressively reduces feature extraction performance, primarily attributed to the introduction of irrelevant or noisy spectral components that interfere with discriminative feature learning. This phenomenon confirms the necessity of balanced sparsity control for maintaining both computational efficiency and representation fidelity.

IV. CONCLUSION

In this article, the MSLTD is proposed for HTD. The method develops a training sample generation strategy that relies solely on raw image data and a single prior target spectrum, effectively addressing the problem of limited training samples. A multiscale feature lightweight network based on a three-channel Transformer is designed, which incorporates the GSBE and MSST modules. These modules facilitate the capture and fusion of multiresolution intrinsic spectral features and long-range spectral dependencies, thereby effectively reducing redundant information. In addition, the AHC loss function is proposed to dynamically enhance the weights of hard-to-classify samples and strengthen the discrimination between targets and background regions. Experimental results on four publicly available hyperspectral datasets demonstrate that the MSLTD method achieves outstanding performance in detection maps, quantitative AUC scores, and target–background separability. Compared with existing methods, the MSLTD achieves a favorable balance between target saliency and background suppression, showcasing excellent detection performance. Furthermore, ablation experiments further validate the effectiveness of the spectral

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mixture model, the SSA module, and the AHC loss function, as well as the impact of the key parameter k on the performance of the SSA module. Future work will focus on further optimizing the model architecture, exploring advanced data augmentation strategies, and evaluating the model in more practical application environments.

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Ying Cui was born in Harbin, Heilongjiang, China, in 1979. She received the Ph.D. degree in signal and information processing and the Postdoctoral degree in computer science and technology from Harbin Engineering University (HEU), Harbin, in 2006 and 2009, respectively.

In 2005, she joined the Faculty Member with HEU, where she is currently an Associate Professor with the School of Information and Communication Engineering. Her research interests include intelligent signal processing, image processing, and wireless sensor network optimization.

Dr. Cui is a reviewer for many journals, including the *Journal of Wuhan University*, *Journal of Harbin Engineering University*, *Journal of Applied Remote Sensing*, *Acta Optica Sinica*, and *Laser and Optoelectronics Progress*.



Tao Chen received the B.E. degree in communication engineering from Guilin University of Electronic Technology, Guilin, China, in 2023. She is currently pursuing the M.S. degree in electronic information with Harbin Engineering University, Harbin, Heilongjiang, China.

Her research interests include deep learning, computer vision, and remote sensing image processing.



Liguo Wang (Member, IEEE) received the M.S. degree in natural science and the Ph.D. degree in engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2002 and 2006, respectively, and the Post-Doctoral degree from Harbin Engineering University, Harbin, in 2009.

He is currently a Professor with the School of Information and Communication Engineering, Dalian Minzu University, Dalian, China.

Dr. Wang is currently a Senior Member of the Chinese Institute of Electronics and China Institute of Communications (CIC) and a member of the Chinese Society for Optical Engineering. He is also a fellow of the International Society for Digital Earth (ISDE). He is also a member of the Chinese Society of Optical Engineering and a Senior Committee Member of the Photoelectric Technology of Astronautics Society. He is a reviewer for many journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Journal of Electronics and Information Technology*.



Shan Gao was born in Yichun, Heilongjiang, China, in 1986. He received the Ph.D. degree in electrical engineering from the Harbin Institute of Technology, Harbin, China, in 2017.

In 2017, he joined the Faculty Member with Harbin Engineering University, Harbin, where he is currently a Lecturer with the College of Information and Communication Engineering. His research interests include artificial intelligence, image processing, and detection technology.



Liwei Chen was born in 1974. She received the Ph.D. degree in signal and information processing from Harbin Engineering University (HEU), Harbin, China, in 2005.

She is currently an Associate Professor with the School of Information and Communication Engineering, HEU. Her research interests include artificial intelligence, deep learning, image segmentation, feature extraction, and classification recognition and understanding.

Dr. Chen is the Editorial Board Member of *Applied Science and Technology*.



Chunhui Zhao received the B.S. and M.S. degrees from Harbin Engineering University, Harbin, China, in 1986 and 1989, respectively, and the Ph.D. degree from the Department of Automatic Measure and Control, Harbin Institute of Technology, Harbin, in 1998.

He was a Post-Doctoral Research Fellow with the College of Underwater Acoustical Engineering, Harbin Engineering University, where he is currently with the College of Information and Communication Engineering, as a Professor and a Ph.D. Supervisor. His research interests include digital signal and image processing, mathematical morphology, and hyperspectral remote sensing image processing.

Dr. Zhao is a Senior Member of the Chinese Electronics Academy.

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